

MEDHA RAKESH

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PROFESSIONAL SUMMARY

UC Berkeley Data Science and Computer Science student (B.A., May 2028) with a strong CS foundation (data structures, algorithms, computer architecture) and a track record of building software end-to-end. Fluent in Python, Java, and C, and full-stack with FastAPI, Next.js, and TypeScript. Built a full-stack platform that placed Top 8 of ~300 teams and an AI incident-response agent, with a bias toward correct, well-tested code over quick demos.

EDUCATION

University of California, Berkeley

Expected Graduation: May 2028

Bachelor of Arts in Data Science, Computer Science

Relevant Coursework: Foundations of Data Science, Structure and Interpretation of Computer Programs, Data Structures, Computer Architecture, Calculus II, Linear Algebra & Differential Equations

WORK EXPERIENCE

The SALT Research Group

January - May 2026

Machine Learning Research Assistant

Berkeley, CA

- Engineered an automated data processing pipeline that parses raw instrument output files, computes statistics across experimental trials, and exports structured data for downstream analysis, replacing manual workflow
- Implemented signal preprocessing logic (frequency filtering, sign-convention correction, and trial averaging) to measure the electrical conductivity of molten salt across a 500–700°C range for nuclear reactor safety monitoring

DataGood at Berkeley

January 2026 - Present

Data Science Researcher

Berkeley, CA

- Built and compared multiple machine learning models (linear regression, random forest, neural network) to predict urban heat island intensity across California census tracts, finding that adding physical environmental features (tree canopy, traffic, ozone) raised model performance from $R^2=0.32$ to $R^2=0.81$
- Identified 357 dual-burden census tracts across LA, Riverside, and San Diego presented analysis and geospatial visualizations through final pitch deck, translating Tableau findings to communicate heat mitigation priorities to non-technical policy partners

Bloomberg

July 2026

KWK Instructor

San Francisco, CA

- Taught SQL, relational database design, and statistical analysis to 30 high school scholars over a two-week intensive, leading labs on multi-table joins and query optimization while providing one-on-one debugging support
- Led capstone project teams through the full software development lifecycle, from requirements gathering and user research through wireframing, iterative code review, and final demo presentations to industry mentors

PROJECTS

ContractAlpha – Government Contract Investment Platform

Top 8 Finalist of ~300 teams | Python, FastAPI, Next.js, TypeScript, REST APIs

- Owned a full-stack platform end-to-end that turns federal contract data into stock investment signals: FastAPI backend, Next.js frontend, and a pipeline ingesting USAspending.gov awards. Engineered a date-cutoff layer preventing lookahead bias in backtests and parallelized batched LLM calls to cut runtime by 20s; the portfolio outperformed the S&P 500 by 24.7%. github.com/Manya-msk/contract-alpha

Cloudscape – AI Incident Commander

Best Use of Perplexity Agent API | Next.js, AWS SDK, Perplexity Sonar, Supabase

- Built an AI-powered tool that autonomously pulls CloudWatch metrics, logs, and CloudTrail events in parallel when an alarm fires, eliminating 20- 30 minutes of manual context-gathering, then returns a likely root cause with a confidence score, ranked fix commands, and a ready-to-use postmortem. github.com/anikag35/cloudscape

Multi Class Risk Identifier

Most Creative Award Datathon for Social Good | Python, Random Forest, Scikit-Learn

- Built a predictive at-home health assessment model using self-measurable data to reduce unnecessary ER visits, earning the Most Creative Award among 30+ teams. github.com/medha-rakesh/DatathonSubmission

SKILLS

Languages: Python, Java, C, TypeScript, JavaScript, SQL, R, HTML/CSS

Frameworks & Tools: FastAPI, Next.js, React Native, REST APIs, scikit-learn, pandas, NumPy, Git, Supabase

Concepts: Data structures, algorithms, full-stack development, data pipelines, machine learning, computer architecture